An overview of gRPC

gRPC is an open source framework for general-purpose Remote Procedure Calls across the network. gRPC is mostly aligned with HTTP 2 semantics, and also allows full-duplex streaming in contrast. It supports different media formats like **Protobuf** (default), JSON, XML, Thrift, and more, though **Protocol Buffer** (**Protobuf**) is performance-wise much higher than others.

You may wonder whether *g* stands for Google in gRPC. It sounds logical, since it was initially developed by Google. In fact, *g*'s meaning is changed with every release. *g* stands for *gRPC* in version 1.0, that is, *gRPC* was *gRPC Remote Procedure Call*. We are going to use version 1.17 in this chapter. For this release, **g** stands for **gizmo**, that is, **gizmo Remote Procedure Call** (**gRPC**). You can track all the ***g*** references at https://github.com/grpc/grpc/blob/master/doc/g_stands_for.md.

gRPC is a layered architecture that has the following layers:

- **Stub**: Stub is the topmost layer. Stubs are generated from the **Interface Definition Language** (or **IDL**) defined file (containing interfaces having service, methods, and messages). By default, Protocol Buffer is used, though you could also use other IDLs like `messagepack`. Client calls server through stubs.
- **Channel:** Channel is a middle layer that provides the **application binary interfaces** (ABIs) that are used by the stubs.
- **Transport:** This is the lowest layer and uses HTTP 2 as its protocol. Therefore, gRPC provides full-duplex communication and multiplex parallel calls over the same network connection.